

Assignment 2 SECPH 2523

QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offences by students in a college.

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
				00-690	15/03/06	1	20.00
				00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
				00-770	19/06/06	2	30.00

a) Table 2.1 is said to be unnormalized table. Why?

Repeating groups and multivalued attributes make this table unnormalized, increasing the risk of data redundancy and inconsistency. For instance, if a single student has several penalties characteristics like StudentID, Name, LicenseNo, and CarReg may be repeatedly stored over several rows. Furthermore, multivalued fields like ReceiptNo, Date, OffenceCode, and Fine (RM) are against the atomicity principles.

b) Transform the unnormalized Table 2.1 into a First Normal Form (1NF) table. Show the transformed 1 NF table. What is the primary key for this new 1 NF table.

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/02/06	2	30.00
F1011	CK Fong	P100	JGK9201	00-690	15/03/06	1	20.00
F1011	CK Fong	P100	JGK9201	00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
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Primary key : studentID and ReceiptNo

QUESTION 2

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
		12	Pauline James	B	\$50
		16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
		17	Monique Williams	B	\$50

1NF:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
1023	Madagascar travel site	12	Pauline James	B	\$50
1023	Madagascar travel site	16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
1056	Online estate agency	17	Monique Williams	B	\$50

Primary Key : Project Number, Employee Number

Foreign Key : none

2NF:

Table 1: Project

Project number	Project name
1023	Madagascar travel site
1056	Online estate agency

Table 2: Employee

Employee number	Employee name	Rate category	Hourly rate
11	Vincent Radebe	A	\$60
12	Pauline James	B	\$50
16	Charles Ramoraz	C	\$40
11	Vincent Radebe	A	\$60
17	Monique Williams	B	\$50

Table 3: Work

Project number	Employee number
1023	11
1023	12
1023	16
1056	11
1056	17

Primary key:

1. Project table: project number
2. Employee table : Employee number
3. Work table : none

Foreign Key:

1. Project table: none
2. Employee table : none
3. Work table : project number + employee number

3NF:

Table 1: Project

Project number	Project name
1023	Madagascar travel site
1056	Online estate agency

Table 2: Employee

Employee number	Employee name
11	Vincent Radebe
12	Pauline James
16	Charles Ramoraz
11	Vincent Radebe
17	Monique Williams

Table 3: Work

Project number	Employee number
1023	11
1023	12
1023	16
1056	11
1056	17

Table 4: Rate

Rate Category	Hourly Rate
A	\$60
B	\$50
C	\$40
A	\$60
B	\$50